

Diego Chuquillanqui

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EDUCATION

Lehigh University, Bethlehem, PA

Graduation: May 2030

Bachelor of Science in Computer Engineering

Relevant Coursework: Integrated Engineering Practice (Sensors, Embedded Systems, Design & Prototyping), Calculus III, Introductory Physics I & Laboratory

Colegio Peruano Norteamericano Abraham Lincoln, Lima, Peru

Graduated: Dec 2025

International Baccalaureate Diploma: 41/45 | Rank: 7/93 | School Honor Guard

GPA: 3.7/4.0

PROFESSIONAL EXPERIENCE

Design Labs Tech Fellow – Electronics Design Studio

Sep 2026 – Present

Lehigh University Design Labs, Bethlehem, PA

- Support student electronics and robotics projects through component selection, circuit prototyping, hardware debugging, and equipment troubleshooting
- Guide students in using microcontrollers, soldering equipment, oscilloscopes, measurement tools, and PCB prototyping resources to build, test, and refine embedded and electromechanical prototypes

Strategy & Business Development Intern

Jan 2026 – Jul 2026

Cuenca & Asociados Asesoría Empresarial Integral, Lima, Peru

- Supported strategy and growth engagements across 5 organizations in mining, healthcare/education, mobility, industrial chemicals, and consumer goods, including new ventures and post-acquisition planning
- Conducted market research, benchmarking, field research, and strategic analysis; translated findings into recommendations, executive presentations, proposals, and business-development materials

TECHNICAL PROJECT EXPERIENCE

Automatic Floodgate Prototype | ENGR 095, Lehigh University

Sep 2026 – Oct 2026

- Designed and built an automatic floodgate in a three-person team using a water sensor and a custom-fabricated PEDOT humidity sensor to detect unsafe moisture conditions and deploy a physical barrier
- Fabricated and characterized the PEDOT sensor and programmed an Arduino Nano ESP32 in MicroPython for sensor data acquisition, threshold detection, and automated stepper-motor control
- Presented the prototype at Lehigh's FYRE Expo as one of two teams selected to represent the module

Musicle – Music Guessing App | React, Vite, Spotify API, Netlify

Jan 2026 – Apr 2026

- Built a Wordle-style daily music guessing web application using the Spotify Web API, featuring daily song selection, saved browser progress, and scalable English and Spanish internationalization
- Secured Spotify authentication through Netlify serverless functions to keep API credentials server-side and implemented catalog filtering and deduplication to remove duplicate and alternate track versions

Gesture-Controlled Robotic Arm | Personal Project, Lehigh Design Labs

Sep 2026 – Present

- Developing a multi-DOF robotic arm designed to reproduce wearable hand/arm gestures in real time using flex and IMU sensing, Arduino-based embedded control, and hybrid stepper/servo actuation
- Integrating 3D-printed mechanical joints, wireless sensor communication, motion filtering, and kinematic control for coordinated multi-axis movement

LEADERSHIP & COMMUNITY INVOLVEMENT

School House Captain, Allen House | Abraham Lincoln School, Lima, Peru

Dec 2024 – Dec 2025

- Led 400+ students across grade levels and coordinated students, parents, and teachers across athletic and creative competitions, helping lead the House to a 1st-place overall finish

Volunteer Group Leader | Nuevo Imperial, Quilmaná, Peru

Mar 2025 – Oct 2025

- Led a 10-person volunteer team in constructing 2 houses for donation to local families, organizing fundraising efforts, coordinating materials, and executing project plans

TECHNICAL SKILLS

Programming & Web: Python, MicroPython, C, JavaScript, HTML/CSS, SQL

Embedded & Hardware: Arduino Uno, Nano ESP32, microcontrollers, sensors, stepper/servo motors, motor drivers, oscilloscopes, circuit/PCB prototyping

Tools: Git/GitHub, Microsoft Excel

Languages: English (Fluent), Spanish (Native)